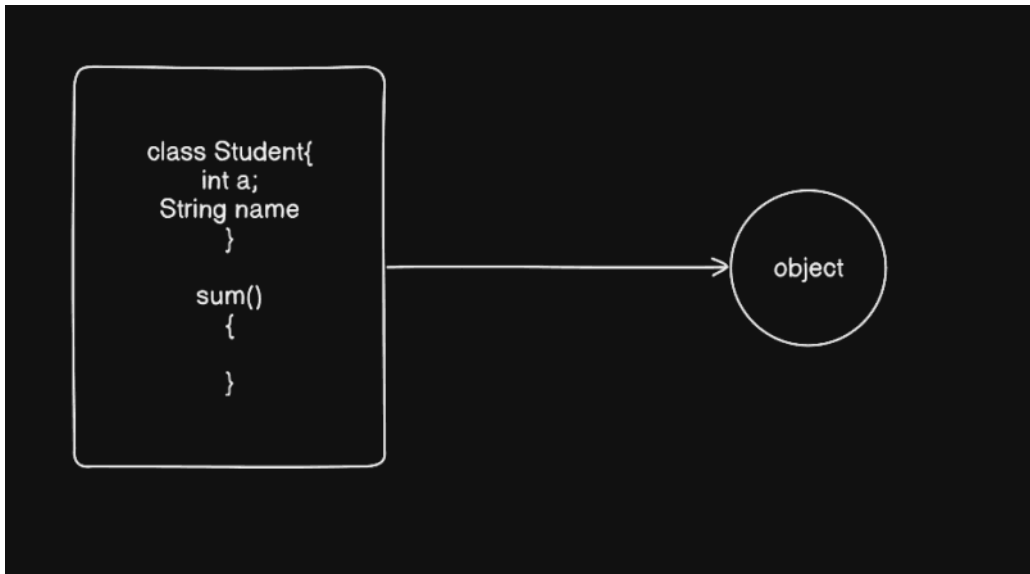


OOPs

27 June 2026 11:05

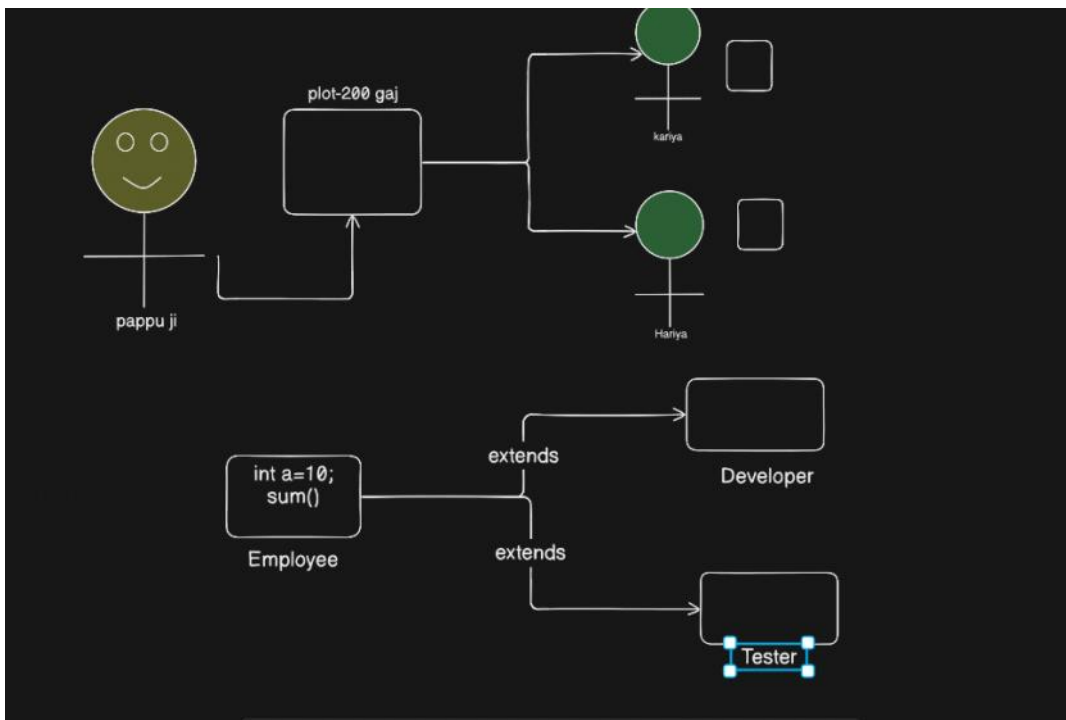
What is OOPs ?
Object oriented programming



What is class ?
What is object

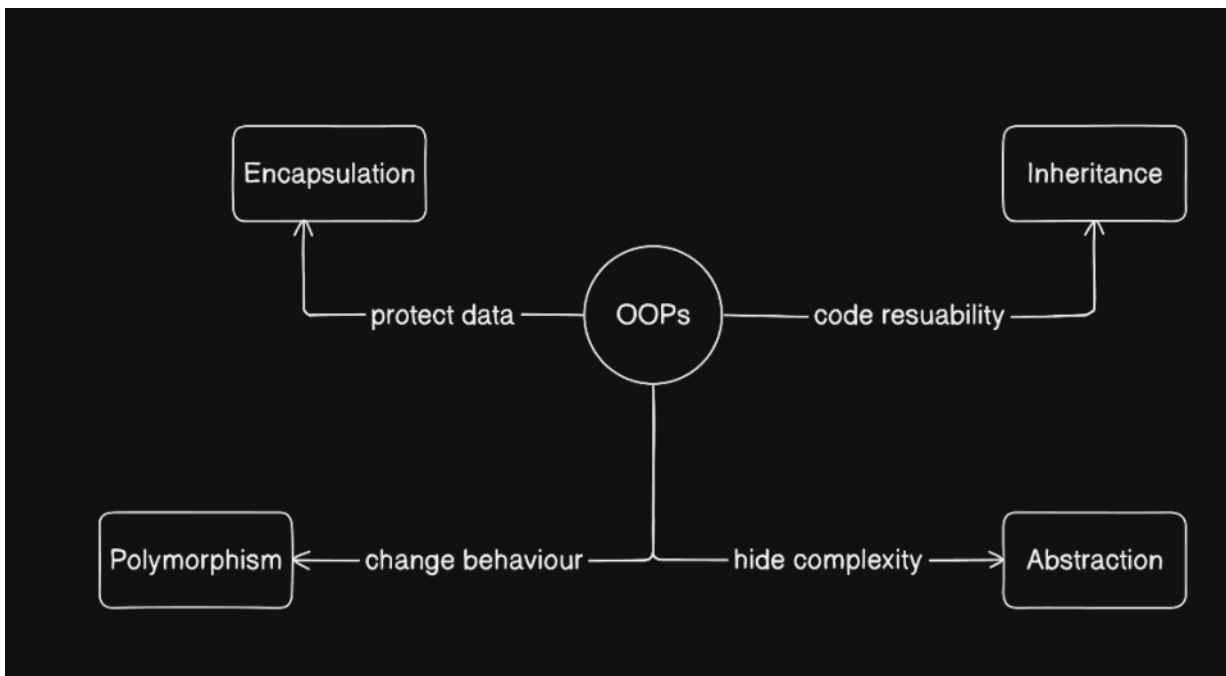
If I say Employee, Manager, Tester, Developer are all of them completely different ?
No. These all are employees

Note: In java common things can be placed in parent class. Specific things can be placed in child class



There are 4 principles in OOPs

1. Inheritance
2. Encapsulation
3. Polymorphism
4. Abstraction



2. Encapsulation

Encapsulation means binding data and methods together inside a class

Encapsulation = data hiding + controlled access

Eg:

ATM machine does not give direct access to bank database

We use buttons and screen options that is controlled access.

```

public class Test {

    public static void main(String[] args) {
        BankAccount bankAccount= new BankAccount();

        bankAccount.deposit( amount: 5000);

        System.out.println(bankAccount.getBalance());
    }
}

```

Balance is private

Outside class cannot directly change balance, money can be added only through deposit().

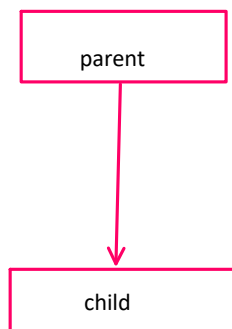
Inheritance:

Child class get properties and behaviour from parent class

Will use extends keyword for make child-parent relationship

Type of inheritance

1. Single inheritance : one parent , one child

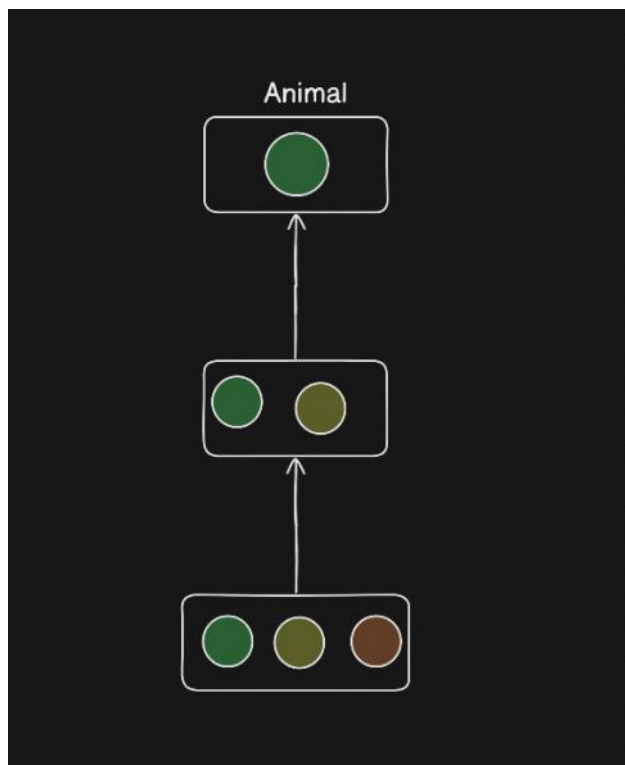


2. Multilevel inheritance

Parent to child to grandchild

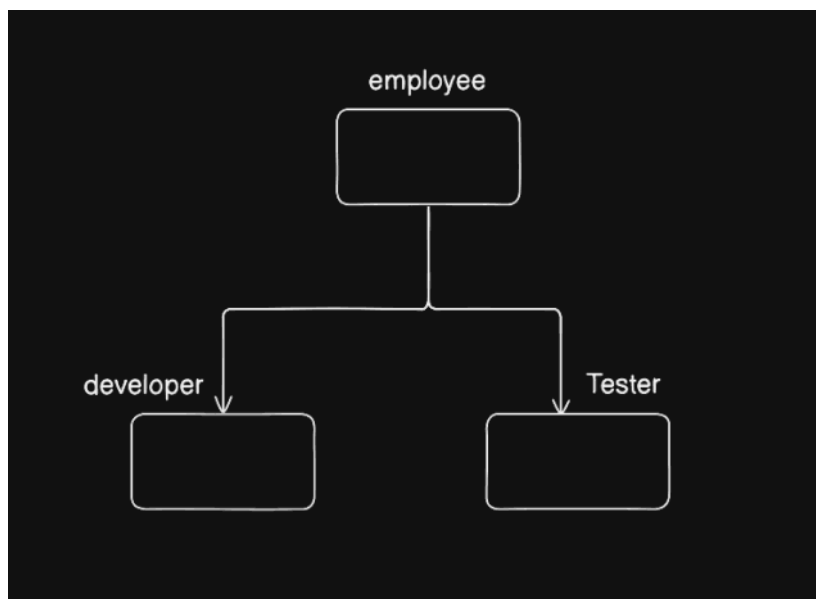
Animal -----> Dog

Dog----->puppy



3. Hierarchical inheritance

One parent, multiple children



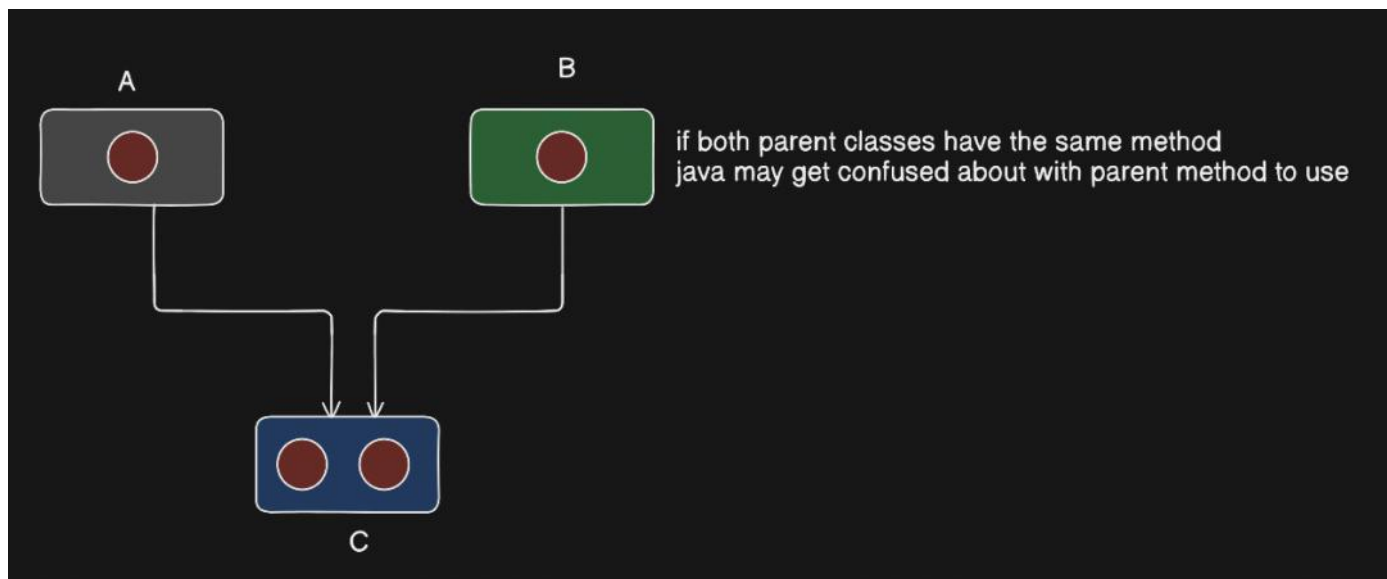
4. Multiple inheritance with classes is not allowed

Java does not allow

Class C extends A, B,

{

}



What is this keyword ?

this keyword is reference variable that refer to current object

Simple :

This means the current object on which the code is running.

Simple:

When one object calls a method, inside that method this points to that same object.



Why we need this ?

When local variable name and instance variable name are same this is used to refer to the instance variable

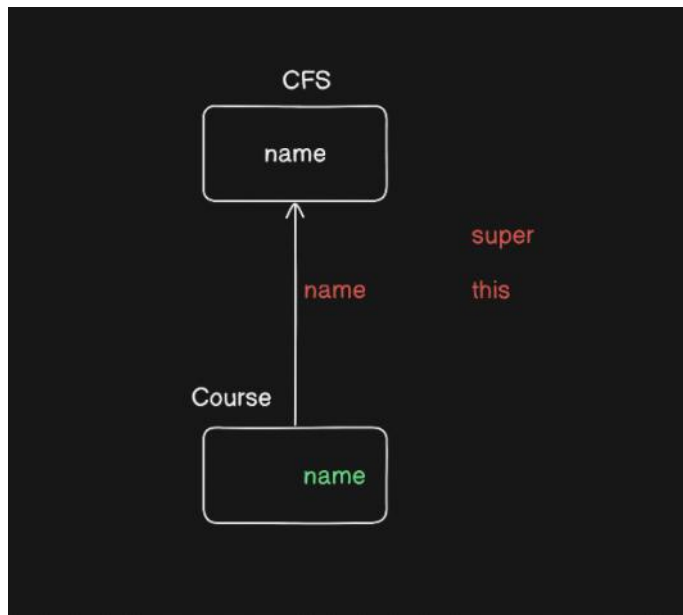
Uses of this:

1. To access current class variable

2. To call current class method
3. To call current class constructor
4. To pass current object as an argument
5. To return current object

What is super keyword ?

Super is reference variable used inside a child class.
Super refer to the immediate parent class part of the current object

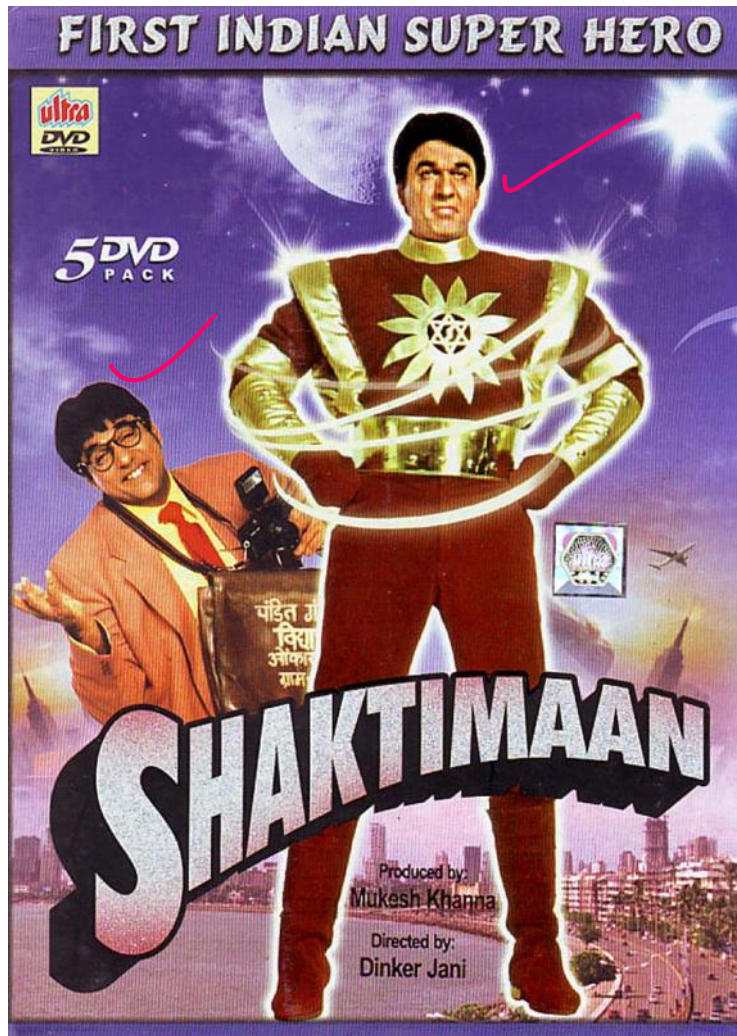


Polymorphism

Polymorphism means one thing can behave in multiple forms

Polymorphism = same name, different behaviour





Real world examples

The word "pay" can mean

1. Pay by cash
2. Pay by upi
3. Pay by card

In java polymorphism mainly have two types

1. Method Overloading
2. Method overriding

Method overlading

Same method name, different parameter, same class.



```
class Calculator no usages
{
    int add(int a, int b) no usages
    {
        return a+b;
    }

    int add(int a, int b, int c) no usages
    {
        return a+b+c;
    }

    double add(double a, double b) no usages
    {
        return a+b;
    }
}
```

Method overriding

Same method name, same parameter, parent -child relationship.


```

2
3 class Payment 1 usage 1 inheritor
4 {
5     void pay() 1 usage
6     {
7         System.out.println("Generic Payment");
8     }
9 }
10
11 class UpiPayment extends Payment 2 usages
12 {
13     @Override 2 usages
14     void pay() {
15         System.out.println("Payment done using UPI");
16     }
17 }
18
19 public class PaymentTest {
20     public static void main(String[] args) {
21         UpiPayment upiPayment = new UpiPayment();
22         upiPayment.pay();
23     }
24 }

```

Dynamic Method Dispatch:

Runtime polymorphism :

Compile time polymorphism

Abstraction

Abstraction means showing necessary details and hiding internal complexity